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1 Introduction

The Tile Calorimeter (TileCal) is a sampling calorimeter of the ATLAS Experiment [1]. TileCal is in charge of identifying and measuring hadronic jets and transverse missing energy. As depicted in Figure 1, the electronics for the read-out of all the TileCal are divided in four cylindrical barrels, each composed of 64 wedge-shaped modules where plastic scintillator tiles are used as active material interleaved with steel plates used as absorbers. Light from two sides of a pseudo-projective cell composed by a group of scintillators is collected by wavelength shifting fibers and read out by two photomultiplier tubes (PMTs). To face the higher radiation levels and increased rates of pileup at the High Luminosity Large Hadron Collider (HL-LHC), the TileCal electronics will be upgraded with a new design that will provide continuous digital read-out of all the TileCal with improved timing and better energy resolution.

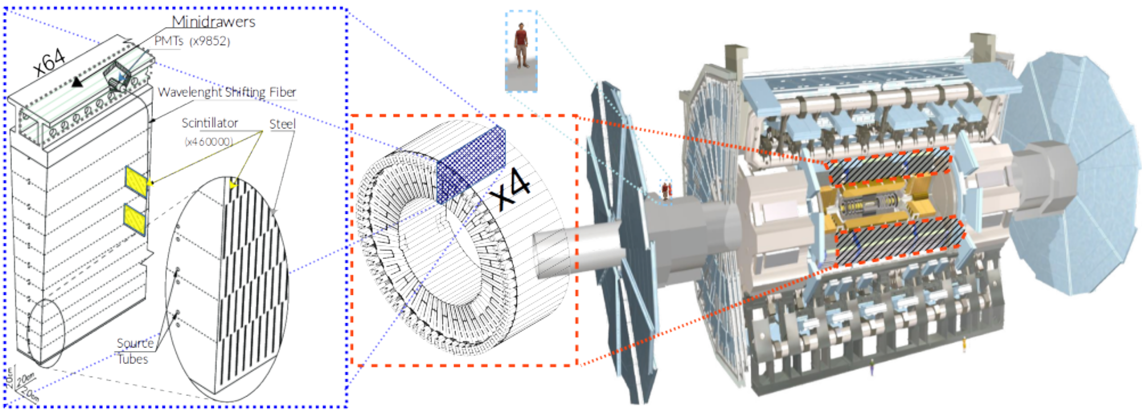

Figure 1. TileCal structure in ATLAS.